

# **BORDER OPTICS — Executive Summary**

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### **A Satellite-Based Verification Framework for India's Vibrant Villages Programme (VVP-I)**

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## **Project Overview**

The origins of BORDER OPTICS lie in one line in a Parliamentary reply. Responding directly to the question, the Ministry of Home Affairs said that no impact assessment has ever been conducted for the Vibrant Villages Programme — this despite the scheme having already sanctioned roughly Rs. 4,800 crore across 2,967 villages, covering five Himalayan border states and union territories from Arunachal Pradesh to Ladakh. There's an intersection of three fields that I don't want to consider separately — the gap that I'm digging to fill. It starts with political science, given that VVP is a textbook example of infrastructure that does double duty as both civilian welfare and geopolitical signaling. The next is economics, a large-scale public investment that has never been costed against the results it delivers. And the third is geospatial science, because satellite imagery lets the Government's implicit claim of development be directly tested instead of taken on faith. So, I tested it — 258 individually geocoded villages benchmarked against a district-restricted non-VVP control group of 732 villages in the same districts, using two equally defensible ways of compositing the same satellite record, checked against each other, against a control group, across a third year, and across three buffer sizes. Two earlier versions of this project each reported a headline finding that turned out, on closer inspection, to be a data artifact rather than a real effect — and both times the right call was to fix the artifact and report what was actually left, not to keep the result. The first was an unresolved ambiguity between the two compositing windows — one showing a significant rise in NDBI, the other none — that depended on an incomplete satellite extraction missing an entire state's worth of villages; once the extraction was completed and the treated and control data were pulled the same day, that “surviving” signal did not survive being recomputed on the complete sample. The second was a night-lights control-group gap that then took its place as the study's one

remaining significant result, until I found that 23 of the 735 “non-VVP” control villages it was computed against were, in fact, the same physical settlements as treated villages counted a second time under a different name — once that contamination was actually removed and the control list regenerated (732 villages, independently verified at zero overlap), that result went null too. What’s left, checked every way this project could check it, is a clean null: no satellite-detectable effect of VVP-I on built-up area or night-lights, in either compositing window, against the control group, or at any of three buffer radii. If cost translated to results, ten times the money should have bought ten times the outcome — it didn’t, in either version of this analysis. Parliament said no one had been looking. This is what looking actually turns up, including the part where two encouraging-looking results, in turn, each had to be walked back once the data behind them was actually fixed.

## **The Question**

When directly asked in a Lok Sabha Unstarred Question (Lok Sabha Question No. 508, 3 February 2026) whether any impact assessment had been done of the Vibrant Villages Programme, the Ministry of Home Affairs responded, “No impact assessment has been carried out.” The question this project set out to answer traces back to that gap: did India’s Rs. 4,800 crore border-village development scheme produce measurable physical development on the ground — and can the answer be trusted, or does it depend on which measurement choice you make?

## **The Method**

258 individually geocoded villages, benchmarked against a district-restricted non-VVP control group of 732 villages in the same 14 districts, were tested across 5 Himalayan border states and UTs. The satellite record was provided by Sentinel-2 built-up-area change (NDBI) and VIIRS night-lights, 2021–2025. In each case, two independent compositing windows were used, specifically full-year and summer-matched. The before/after change was addressed by Wilcoxon signed-rank tests, the budget/border-proximity relationships were addressed by Spearman correlations, and the comparison with the control group was addressed using a district fixed-effects Difference-in-Differences model. Two more checks are added to the core design: the 2021/2025 comparison is expanded to a 3-point 2021/2023/2025 trend, and the fixed extraction buffer of 500m is compared to alternative extraction buffers of 250m and 1km.

## **The Finding**

The built-up-area result initially looked compositing-window sensitive: no significant change in a full-year composite, but a highly significant change in the same villages recomposited to a June–September summer-matched window. That summer-window result turned out to depend on an incomplete extraction — at the time it was run, all 31 Sikkim villages returned zero usable summer imagery, a gap since traced to the Sentinel-2 archive still backfilling scenes for those dates rather than to any real absence of change (Development Log Entries 21–22). Once the extraction was completed — same method, same dates, pulled the same day as the control group’s own data — the summer-window result reversed to a null, matching the full-year window. Every check that had been run against the old result was rerun against the complete one, and every one of them now agrees: the summer NDBI gap against the district-restricted control group is no longer significant ( $p = 0.447$ , was  $p = 0.00033$ ), it no longer holds

at any of the three buffer radii tested (all null now, were all significant), it's no longer significant with any one of the 14 control-group districts dropped (0 of 14 reruns, was 14 of 14), and a randomization-inference test that doesn't lean on cluster asymptotics agrees ( $p = 0.115$ , was  $p = 0.0005$ ). One new result appeared in the same complete data: built-up change now correlates with distance to the border/LAC in the summer window (in the opposite direction from what the securitization hypothesis predicted) — flagged here as a fresh, open finding that hasn't yet been through the same robustness checks, not folded into the numbers below.

| Test                                 | Full-Year Window                                                                                                                                       | Summer-Matched Window                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| NDBI change (Wilcoxon)               | $p = 1.000$ — not significant                                                                                                                          | $p = 0.436$ — not significant (was $p < 0.000001$ on an incomplete sample)                       |
| Night-lights change (Wilcoxon)       | $p = 0.050$                                                                                                                                            | $p = 0.9999$                                                                                     |
| Control-group DiD (NDBI, H4)         | $p = 0.310$ — not significant                                                                                                                          | $p = 0.473$ — not significant (was $p = 0.00033$ )                                               |
| Control-group DiD (Night-lights, H4) | $p = 0.309$ (FE) / $p = 0.578$ (no-FE) — not significant (was $p = 0.036$ FE / $p = 0.103$ no-FE on the contaminated control list; resolved, Entry 25) | $p = 0.450$ — not significant (was $p = 0.0044$ )                                                |
| Budget vs. measured change           | ~10x budget gap (AP vs. UK) → smaller mean change for the larger budget, $n=3$ states, not powered                                                     | same direction, $n=3$ states, not powered — sign flips relative to full-year at this sample size |

No control-group DiD result, for either outcome or window, clears significance under any specification any more. The full-year night-lights gap had, since an earlier version of this project, been the one exception — but it turned out to rest on a contaminated control group: 20 of the old 735 control villages were physically the same settlements as treated villages under a different name spelling or script, and 3 more were officially-listed VVP-I villages under their own real, ungeocoded names. Once that contamination was actually removed and the control list regenerated (732 villages, verified at zero overlap — Development Log Entry 25), that gap went null on every specification that had previously found it significant: the primary estimate ( $p = 0.036 \rightarrow p = 0.309$ ), the HC3 comparison ( $p = 0.103 \rightarrow p = 0.578$ ), leave-one-district-out (10 of 14 significant  $\rightarrow$  0 of 14), and randomization inference ( $p = 0.0155 \rightarrow p = 0.130$ ). Budget tells the same story either way: Arunachal Pradesh's sanctioned budget of Rs. 2,749.74 crore comes close to ten times Uttarakhand's Rs. 270.58 crore, and the two states' average measured built-up change doesn't track that gap in either compositing window — now checkable against all three core states instead of two, since Sikkim's data is no longer missing.

## Validation & Robustness Checklist

- ✓ Dual compositing-window test (full-year + summer-matched) — now agree with each other on NDBI, both null
- ✓ Two independent metrics tested separately (NDBI + VIIRS)
- ✓ District-restricted non-VVP control group (732 villages, same 14

districts, DiD design) — fully populated, no missing rows on either side, independently verified free of treated-village contamination (Development Log Entry 25)

- ✓ Control-group DiD checked further: leave-one-district-out and randomization inference, rerun on the complete, decontaminated data — 0/14 significant in every one of the four outcome/window combinations (was 10/14 for full-year night-lights before the contamination fix)
- ✓ Three-point multi-year trend (2021/2023/2025), not a single before/after pair — unaffected by this update, consistent with the now-null two-point comparison
- ✓ Buffer-radius sensitivity swept (250m/500m/1km) — all three now null and in agreement
- ✓ Cross-checked against sanctioned budget (Parliamentary record) — now across all three core states, not two
- ✓ Every data gap disclosed (Ladakh, HP, Uttarakhand caveats)
- ✓ The original “surviving” summer NDBI signal was itself checked once the extraction was completed — and did not survive. Disclosed and superseded rather than left standing (Development Log Entries 21–22).
- ✓ The full-year night-lights control-group gap that took its place was itself checked once the control-group contamination was found — and it did not survive either. Disclosed and superseded, not left standing (Development Log Entries 23–25).
- △ A new summer-window border-proximity correlation (H3) has emerged in the complete data and has not yet been through the same robustness checks as the two results it replaces

## Honest Limitation

An earlier version of this project reported the built-up-area result as a compositing-window instability worth taking seriously — no significant change in a full-year composite, a highly significant one in the same villages recomposited to summer. That summer result depended on an incomplete extraction: monsoon cloud cover had, at the time it was run, wiped out usable summer imagery for all 31 Sikkim villages, and the resulting 154-village sample was not a random subset of the 251-village core sample — it differed from the 97 villages that dropped out both in state composition and in 2021 baseline NDBI ( $p = 0.00063$ ). What wasn't clear at the time was whether Sikkim's “zero valid data” was a permanent fact or a snapshot of how far the Sentinel-2 archive had backfilled scenes on the day the extraction ran. It was the latter. A complete re-extraction, pulled the same day as the control group's own data, recovered real values for all 31 Sikkim villages — and the summer-window result, once computed on the complete sample, was a null, not a significant increase. Every downstream check built around the old result (the control-group DiD, the buffer-radius sweep, leave-one-district-out, randomization inference) was rerun on the complete data and now agrees: null. The full-year night-lights gap then took that result's place as the one control-group result still significant under any specification — but it, too, turned out to depend on a data problem rather than a real effect: 20 of the 735 control villages it was tested against were physically the same settlements as treated villages under a different name spelling or script, and 3 more were officially-listed VVP-I villages under their own real, ungeocoded names, so some of what looked like a genuine treated-vs-control gap was really the treated group being compared, in part, against itself. Once that contamination was actually removed — the control list regenerated at 732 villages, independently verified at zero coordinate overlap and zero name matches — the gap went null on every specification that had previously found it significant, including the leave-one-district-out check (0 of 14 significant, was 10 of 14) and randomization inference ( $p = 0.130$ , was  $p = 0.0155$ ). A new correlation between built-up change and border proximity has appeared in the same complete summer data and is flagged as an

open question rather than a confirmed one, for exactly the reason both of the old results above should have been flagged before being reported as this study's headline. Due to limitations on sample size and geometry, the border-proximity correlation generally (H3) is still treated as exploratory. The data gaps — the “unresolved” villages in Ladakh, the “partial” sample in HP, and the “block” confusion in Uttarakhand — are not glossed over here, and neither is the fact that this project's own first two reported headline results, in turn, did not hold up.

## Real-World Relevance

Parliament was told no independent impact study exists for a Rs. 4,800 crore government scheme spanning 2,967 sanctioned villages — this project is that study, including the part where its own first answer had to be corrected. It demonstrates the feasibility of satellite methods for verifying public infrastructure projects, and it demonstrates something else, arguably more important for how this kind of study should be done: a result that passes every robustness check thrown at it — a control group, a buffer-radius sweep, a multi-year extension, a multiple-testing correction — can still be wrong if the underlying extraction was incomplete, and the only way to find that out is to keep checking the extraction itself, not just the statistics built on top of it. The useful answer for a policymaker isn't to trust whichever result survives the most tests; it's to ask whether the underlying data was actually complete before treating any test's outcome as settled. Once this study's own data was complete, no satellite-detectable effect of VVP-I survived — a real, reportable answer to a question Parliament said had never been asked, and a caution about how easily an incomplete extraction can look like a robust finding until someone checks.

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GitHub: [github.com/sakshimaske303-commits/BORDER\\_OPTICS](https://github.com/sakshimaske303-commits/BORDER_OPTICS) | Live Dashboard: [borderoptics-bkx3lpcvfghdpa2hmuqwsq.streamlit.app](https://borderoptics-bkx3lpcvfghdpa2hmuqwsq.streamlit.app) | Zenodo DOI: [10.5281/zenodo.21759970](https://doi.org/10.5281/zenodo.21759970)

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